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How Do They Decide? Performance Support Staff Perceptions of Decision-Making in Elite-Sport Environments

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Abstract

Wilson, PJ, Roe, G, and Kiely, J. How do they decide? Performance support staff perceptions of decision-making in elite-sport environments. *J Strength Cond Res* XX(X): 000–000, 2024—Decision-making is a critical component of performance support staff responsibilities, yet research exploring staff perceptions relating to key facets of decision-making is lacking. This study aimed to explore perceptions of performance support staff relating to; the importance of decision-making in their role, the value of team decision-making, strategies to enhance decision-making skills, monitoring the process and outcome of decisions, and the obstacles to good decision-making within their professional contexts. A mixed-methods approached was used to explore perceptions of decision-making in 191 performance support staff (male staff = 179, female staff = 12) working in professional (n = 147) or international (n = 44) sport. Performance support staff reported decision-making as extremely important to their role; however, only 31% had received formal decision-making education and revealed *underdeveloped decision-making capabilities* within their organizations as a barrier to good decision-making. Most performance support staff conveyed that team decision-making is a valuable endeavor and augments decision outcomes; yet articulated *dysfunctional team dynamics* as an obstacle to good decision-making in their environments. Largely, performance support staff reported an absence of organizational strategy and *workflow systems and processes* for tracking and making decisions, respectively, whereas only 22% had a consistent reviewing and updating process. When making decisions, performance support staff faced several different *cognitive biases* and were challenged by the *environmental pressures*. The perspectives documented here suggest that performance support staff would likely benefit from undertaking evidence-based training and implementing scientifically informed strategies to enhance decision-making processes within their organizations.

Key Words: teaming, training, tracking, professional sport, judgement and decision-making, barriers

Introduction

Decision-making is a critical component of performance support staff responsibilities (17,47) and involves choosing a particular course of action from a set of alternative options (38). In consequential circumstances, the outcome of each option must be carefully considered and forecasted as accurately as possible (17), before deciding on a course of action. To do so, performance support staff must contend with their innate human shortcomings (45). Specifically, human cognitive limitations restrict information processing, particularly under time constraints (38), resulting in a reliance on heuristics (intuitive mental shortcuts) to simplify the decision-making process (35). However, when used inappropriately, heuristics can lead to cognitive biases (44), which can negatively affect decision accuracy (1,5,12,16).

In addition, performance support staff must also consider the validity of the context in which a decision is being made (45). Validity here refers to the nature of the relationship between information gleaned from the environment by the decision maker (cues) and decision outcomes (21). In high-validity contexts, consistent patterns of behavior provide opportunity for repeated exposure to similar cases, and relevant cues have high predictive power (21,37). For example, early-stage injury rehabilitation is primarily focused on facilitating healing, moderating pain, and reestablishing physical capacity in controlled, and often predictable, circumstances (24). In such circumstances, information such as pain, strength, and biomechanical markers provide clear and measureable indicators of progress (23,25). However, in low-validity contexts, environmental conditions are often volatile, reducing the similarity between cases and the predictive usefulness of cues (21,37). Thus, because of increased uncertainty, forecasting the outcomes of decision options becomes more challenging (45). Continuing the previous example, in the latter stages of rehabilitation when athletes are nearing return to play, practitioners are often faced with the complex (and potentially impossible) task of quantifying training dose-response relationships (18) in the absence of appropriately validated measures (14). Furthermore, the resulting noisy and often incomplete information (36) is interpreted and rationalized by multiple stakeholders with differing experiences, education, and competing agendas (47).

In light of the challenges often associated with decision-making, researchers recently set out to identify the traits and characteristics of high-performing forecasters (4,8,28). Positive

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findings from multiple studies led to the development of evidence-based strategies for enhancing decision-making, in high- and low-validy contexts (42). These include developing the skills to work within a team, implementing rigorous monitoring of the process and outcome of decisions, and training to mitigate against common cognitive biases (9,29,42). Given the importance of decision-making in performance support staff's roles, such strategies may be of benefit. Indeed, recent research has suggested that evidence-based decision-making models and strategies should be adopted in applied sport practice to enhance decision quality (48). However, research investigating if and how evidence-based decision-making strategies are implemented by performance support staff is currently lacking. Research in this area would help identify potential opportunities for improving decision-making in this cohort.

Accordingly, the overarching aim of the present study was to investigate performance support staff's perceptions of decision-making within their professional contexts and explore their use of evidence-based decision-making strategies by focusing on the following objectives:

- Determine performance support staff's perceptions of the importance of decision-making in their roles.
- Investigate performance support staff's perceptions of the benefits of making decisions individually or in teams and of training strategies to enhance decision-making.
- Explore how performance support staff monitor the process and outcome of decisions and determine the obstacles to good decision-making that performance support staff face.

Methods

Experimental Approach to the Problem

The current study was philosophically underpinned by pragmatism, meaning that is was primarily undertaken with the intention of producing practically meaningful knowledge (15). Rather than getting caught up in ontological and epistemological debates associated with traditional research paradigms (e.g., positivism and constructivism), pragmatism asserts that research should be focused on generating useful knowledge that provides practical solutions to an a priori identified problem (32). In the present study, we aimed to understand performance support staff's perceptions of different facets of decision-making to ultimately determine if and where decision-making processes in sport contexts may be enhanced. To do so, a pragmatic process of inquiry was implemented by the authors, whereby the methods used were perceived to be the most effective for addressing the research aims (31). Specifically, a quantitative approach was used to measure and summarize performance support staff's responses to specific statements (closed-ended questions) relating to the research aims. Where subjects' opinions were sought regarding a broader topic (obstacles to decision-making, open-ended question), a qualitative method (thematic analysis) was implemented as a process of identifying patterns within the answers the subjects provided.

This exploratory study used a mixed-methods design, using a questionnaire to collect quantitative and qualitative information on performance support staff's perceptions of decision-making in their professional contexts. Ethical approval was obtained from the Education and Health Sciences Research Ethics Committee at the University of Limerick, Ireland. Consent was obtained from all subjects before filling out the questionnaire.

Subjects

Nonprobabilistic sampling was used to recruit subjects in this study. The study was advertised on social media, through LinkedIn and Twitter, for an 8-week period. In addition, during this period, the authors used a combination of purposive and snowball sampling within their professional networks (authors have a collective of 40 years working in professional sport and a diverse network of professional colleagues). Initially, subjects meeting the inclusion criteria were contacted by the authors. When subjects provided referrals, he or she was also contacted if the inclusion criteria were met. To participate, individuals were required to be (a) a performance support staff member and (b) working in professional or international sport. Two hundred forty responses were collected, with 49 responses excluded for either not meeting the inclusion criteria (*not a performance support staff member and/or* not working in professional or international sport) or had incomplete questionnaires. The final sample included 191 subjects (179 male and 12 female subjects), categorized into professional ($n = 147$) or international ($n = 44$). Support staff identified their experience by less than 5 years' experience ($n = 26$), between 5 and 10 years' experience ($n = 60$) and 10+ years' experience ($n = 105$). Sports included volleyball, rugby, tennis, basketball, cricket, baseball, football (soccer), golf, and hockey. Roles included chief executive officers, performance directors, team managers, head coaches, assistant coaches, sports scientists, athletic performance coaches, strength and conditioning coaches, physiotherapists, performance nutritionists, psychologists, and performance analysts.

Procedures

The research study used an online questionnaire that included both closed- and open-ended questions (see Appendix 1, Supplemental Digital Content, http://links.lww.com/JSCR/A553). In line with previous research that identified strategies used by, and characteristics of, good forecasters (8,27,42), closed-ended questions were designed to investigate if performance support staff favor making decisions individually or in teams, how they perceive training strategies to enhance decision-making and investigate if, and by what means, performance support staff monitor the process and outcome of decisions. An additional closed-ended question was included to determine if performance support staff perceived decision-making as important in their role. Finally, an open-ended question was incorporated to capture performance support staff's perceptions of barriers to good decision-making in their professional environments. After initial design, the questionnaire was sent to a panel of 5 experts (minimum of 5 years of experience in a leadership role in professional sport and with experience of decision-making research) to assess content validity and face validity (3,40). The experts were provided with clear context on the study's aims and detailed explanations of each question. They were asked to assess and provide feedback on whether the questionnaire would effectively capture the facets of decision-making under investigation (content validity) and whether the structure of the questionnaire and the content of each question was clear and understandable (face validity). They were also asked to consider questionnaire length and practitioner burden. In certain instances, members of the panel suggested rephrasing or restructuring questions to better align with the study's objectives. This process continued iteratively until both the expert panel and the authors were confident that the questionnaire was clear, appropriate, and capable of

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capturing relevant information from performance support staff. The final questionnaire consisted of 9 questions on decision-making and 6 demographic questions. Of the 9 questions relating to decision-making, 8 were closed-ended and 1 was open-ended. Three closed-ended questions included agreement Likert scales that ranged from 1 (strongly disagree) to 7 (strongly agree) and 5 closed-ended questions that consisted of single-choice categorical answers (see Appendix 1, Supplemental Digital Content, http://links.lww.com/JSCR/A553).

Statistical Analyses

Statistical analysis was conducted using Qualtrics XM and Microsoft Excel. Incomplete questionnaires, or those completed by subjects not meeting inclusion criteria, were not included in the analysis. Descriptive statistics were calculated in Microsoft Excel (version 16.73) to summarize and report quantitative data from the closed-ended questions. For the categorical questions, the proportion (%) of performance support staff responses to each answer were reported. For the Likert scales, the proportion (%) of responses for each scale value (i.e., 1–7) was provided along with the median value and interquartile range (IQR). Qualitative data analysis was performed on the open-ended question using thematic analysis as described by Braun et al. (6). Key stages in this analysis were familiarization, coding, theme development, refining, naming, and writing up. In each of the first 5 stages (familiarization, coding, theme development, refining, and naming), author 1 (P.J.W.) completed the stage and then presented his findings to author 2 (G.R.), who then critically evaluated the analysis providing feedback. Stages were not progressed until consensus between the authors was met.

Results

Results from the closed-ended questions are presented in Table 1. Decision-making was perceived as important in the role of performance support staff (Table 1A), with most practitioners (98%) rating it as 5 or above (median response 7, IQR 6–7). Most performance support staff perceived collaborative decision-making as valuable, with 97% rating it as 5 or more (median response 6, IQR 5–7). Furthermore, 90% believed that team decision-making yields superior outcomes over individual decision-making (Table 1C). In assessing the preferred path to enhancing decision-making, performance support staff demonstrated a notable inclination toward experiential learning over specific training (median response 3, IQR 3–4), (Table 1D). Thirty-one percent of performance support staff reported having received specific decision-making training (Table 1E). When questioned about specific strategies for tracking the process and outcome of decisions, 69% of practitioners identified that there was no formal strategy in place to "score the value of key decisions and examine how decisions, and perhaps mistakes, were made" (Table 1F). With respect to decision-making philosophies, only 22% of practitioners expressed a commitment to consistently reviewing decisions and persistently adjusting plans accordingly (Table 1G). When asked to describe how they develop decision-making skills, there was a near-to-equal weighting between nonsystematically reflecting as a group (42%) and nonsystematic individual reflection (44%), whereas only 14% reported personally reflecting using a formal, systematic process (Table 1H).

Table 2 details the results from the thematic analysis of responses to the open-ended question "What are the main obstacles to good decision making in your professional environment?." From subjects' responses, 5 themes and 13 subthemes emerged (see Table 2 for supporting theme descriptions and associated supporting data extracts). Themes included *environmental pressures, lack of systems and processes*, *underdeveloped decision-making capabilities*, *dysfunctional team dynamics and cognitive biases*.

Discussion

The survey aimed to explore performance support staff's perceptions of the importance of decision-making in their roles. In addition, it aimed to examine performance support staff's perceptions regarding making decisions individually or in teams and the potential of training strategies to enhance decision-making efficacy. Furthermore, this survey investigated how performance support staff monitor decision-making processes and outcomes, and sought to determine common obstacles to good decision-making. Performance support staff reported decision-making as extremely important to their role; however, only 31% had received formal decision-making education and revealed *underdeveloped decision-making capabilities* within their organizations as a barrier to good decision-making. Regarding decision-making skill development, the majority were slightly more in favor of experiential learning than targeted decision-making training. Most performance support staff conveyed that team decision-making is a valuable endeavor and augments decision outcomes; yet articulated *dysfunctional team dynamics* as an obstacle to good decision-making in their environment. The majority of performance support staff reported an absence of organizational strategy and *systems and processes* for tracking and making decisions respectively, whereas only 22% of practitioners had a consistent reviewing and updating process. Finally, when making decisions, performance support staff faced several different *cognitive biases* and were challenged by the *environmental pressures*. The evidence suggests that performance support staff may benefit from undertaking evidence-based training and implementing scientifically informed strategies to enhance decision-making processes within their organizations.

Results indicated that decision-making is a highly valued skill among the sampled performance support staff (Table 1, median 7 [IQR 6–7]), yet only 31% reported received formal decision-making education. As the questionnaire did not investigate the type of training undertaken, it is not possible to determine if all 31% of performance support staff undertook meaningful decision-making instruction. Results also revealed performance support staff's concerns regarding *unqualified senior management* and a *lack of decision-making awareness and education* within their organizations (Table 2), potentially contributing to the low level of decision-making training in this sample. This highlights the need for recruitment and development processes that target decision-making skills in professional sports organizations in which such skills are lacking.

Notably, when questioned if experience or training was more effective for enhancing decision-making skills, performance support staff in the present study demonstrated an inclination toward experiential learning (Table 1; median 3 [IQR 3–4]). Decision-

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Table 1
Closed-ended questions.

A) How important is decision-making in your professional role?						
Not at all						*Extremely*
1	2	3	4	5	6	7
0%	0%	0%	2%	8%	19%	71%
B) In general, when making difficult decisions (when designing fitness and/or rehab interventions, for example) does consulting with colleagues add meaningful value, i.e., does the time and effort expended in collaboration, justify the added value to the decision?						
Rarely adds value						*Always adds value*
1	2	3	4	5	6	7
0%	0%	0%	3%	24%	30%	43%
C) Which of these statements do you most agree with? When deciding on a course of action within my professional domain						
Collaborative decision making in teams normally results in better outcomes than when I make decisions on my own						90%
Whether I collaborate or make decisions on my own does not typically influence the outcomes of those decisions						7%
Decisions I make on my own normally result in better outcomes than when I make collaborative decisions in teams						3%
D) Is decision-making best improved through experience or specific training in reasoning and thinking skills?						
All experience						*Specific training*
1	2	3	4	5	6	7
4%	13%	36%	31%	10%	4%	2%
E) Have you received any education and/or training interventions directly focussed on enhancing decision-making capabilities?						
Yes						31%
No						69%
F) Within your organisation is there a formal strategy in place to score the value of key decisions and examine how decisions, and perhaps mistakes, were made?						
No, we have no formal strategy in place						69%
Yes, we score the final outcome as positive or negative and evaluate the process						22%
Yes, we score the final outcome as positive or negative but do not evaluate the process						9%
G) Which of these statements best describes your decision-making philosophy? When I think about a planning problem and make a decision,						
I periodically review the initial decision and change the plan accordingly						45%
I consistently review the initial decision and persistently adjust the plan						22%
If I feel my initial decision is no longer valid, I change the plan						18%
I stick with that initial decision unless provided with strong evidence that the plan should change						13%
I stick with that initial decision unless something happens that forces me to change the plan						2%
H) Which of these statements best describes how you have learned to be a better decision-maker?						
I collaboratively reflect through discussion, with groups and/or colleagues (nonsystematically/no process)						42%
I personally reflect, using an informal, nonsystematic process						40%
I personally reflect, using a formal, systematic process						14%
After making a decision I move on to the next problem and learn by experience, through trial and error						4%

making training is not commonplace as a stand-alone component in sport and exercise curricula. Therefore, it is possible that performance support staff were not fully aware of the benefits of, and approaches to, formal decision-making training. Studies suggest that brief interventions can significantly enhance decision-making (8,29), and thus may be appropriate for the busy schedules of performance support staff. Nevertheless, although results from this study suggest that training would be of benefit (e.g., *cognitive biases* being reported as a significant barrier to good decision-making, Table 2, discussed below), further research is required to explore performance support staff's proclivity toward learning by experience over targeted training before recommendations can be made.

Most (90%) sampled performance support staff conveyed that team decision-making augmented decision outcomes (Table 1C) and agreed that consulting with colleagues added meaningful value (median 6 [IQR 5–7], Table 1B). However, performance support staff reported *dysfunctional team dynamics* as a major obstacle to good decision-making, mentioning *poor communication*, *lack of trust*, and *conflict* as contributing factors (Table 2). Research has shown that effective teamwork contributes to better decision outcomes in various contexts (26,29). For instance, Edmondson and Harvey (10) found that teams characterized by high levels of psychological safety, trust, and open communication were more likely to make accurate and effective decisions. The results highlight the potential need for sports organizations to explore strategies to enhance performance support staff's ability to work in teams in environments where such barriers are present.

Notably, only 22% of performance support staff sampled in this study reported consistently reviewing and adjusting plans when making a decision (Table 1G). A significant portion of practitioners (69%) acknowledged the absence of formal strategies for tracking decisions in their organization (Table 1F). Indeed, performance support staff also reported a *lack of systems and processes* as an obstacle to good decision-making (Table 2). The findings of recent research serve to emphasize that the best forecasters, who are known for their exceptional decision-making abilities, consistently and thoughtfully update their forecasts, avoiding confirmation bias while remaining adaptable, and changing direction appropriately as new information presents itself (4,28). As performance support staff are often faced with complex decisions that require regular updates (e.g., when deciding if an injured athlete should compete in the near future (47)), embedding formal strategies for tracking major decisions would likely be of benefit.

In addition to the obstacles discussed in the above sections, results revealed that performance support staff in the present study frequently faced cognitive biases such as *conformity*, *groupthink*, and *outcome bias* (Table 2), indicating the tendency

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Table 2

Themes and associated descriptions, subthemes, and supporting data extracts relating to decision-making obstacles perceived by performance support staff.

Theme	Description	Subtheme	Data extracts
Environmental pressures	Subjects explicitly described specific pressures within their working environments	Competition pressure	"There are competition pressures, that can override what would logically be the correct course of action" "Level of competition, time of season—knockout game" "Result pressure"
		Management pressure	"Pressure from above (going against best judgement to please boss)" "External pressures from Coaches who can 'override' any decision made"
		Time pressure	"Time constraints are a limiter as well; you may not always have the optimal amount of time to sit down and review your situation. It may be an instantaneous decision that is needed" "Time to make decisions is often limited. Often many conflicting factors in play (e.g., long- vs short-term view)"
Lack of systems and processes	Subjects described aspects of the organisational structure within their place of work that were insufficient to support good decision-making	Lack of workflow systems	"Working off a clearer framework to assist in how we make decisions on both a personal level & also in the multidisciplinary teams setting—we don't all operate in the same way which can lead to inconsistencies" "Non-existence of a high-quality knowledge management model, where knowledge, evidence and information quality required for data-informed decision-making processes are optimised to support holistic organisational goals"
		Lack of processes	"Not having a method or process in place to evaluate the value add or impact of our input in any given situation. An absence of this means that we have no way apart from gut feel to measure the impact of our role. If we can't do that then we can't even begin to explore the influence of differing decisions on the similar situations" "Lack of decision-making structure (what's led us to this decision?)" "Lack of formalised review process (was this in/effective?)"
Underdeveloped decision-making capabilities	Subjects referenced deficits in decision-making knowledge and skills of other performance support staff	Unqualified senior management	"We tend to see people get promoted into positions of high responsibilities (who make big decisions each day) on the basis of them being competent practitioners but not necessarily competent leaders or decision makers" "People in wrong roles—head of department in roles because they are most experienced coach not that they are the best manager"
		Lack of decision-making awareness and education	"A lack of education around the area of decision making and leadership" "Decision-making as a skill isn't necessarily emphasized (or understood) in our environment" "No education in decision making"

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Table 2

Themes and associated descriptions, subthemes, and supporting data extracts relating to decision-making obstacles perceived by performance support staff. *(Continued)*

Theme	Description	Subtheme	Data extracts
Dysfunctional team dynamics	Subjects explicated particular behaviours and actions of other practitioners when working in teams	Poor communication	"Under-communication or practitioners not having regular conversations regarding a plan for short-mid-long term and making a decision which affects the training process of an athlete or team WITHOUT consulting those involved even before or after acting"
			"Lack of staff communication"
			"Poor communication"
		Lack of trust	"…a lack of trust/collaboration between support staff"
			"Main obstacles to effective decision making could be … and others trust"
			"… trusting colleagues to help make good decisions"
		Conflict	"Conflict between the club"
			"Multi disciplinary teams, sometimes can't find a middle ground"
			"Conflict between stakeholder success criterion by which decisions are judged (for example a player makes a full recovery from injury by missing games but team does not make play offs)"
			"Conflict of interest from coaches, players and medics—especially over return to play from injury around key games. Having to juggle different opinions from conflicting stand points"
Cognitive biases	Subjects implicitly and explicitly described different cognitive biases that impaired good decision-making	Interpersonal biases; conformity, group think and outcome bias	"Fitting in with department views"
			"Legacy, group think, conflicting goals of others"
			"Previous outcome of similar type decisions which then affect the next decision"
		Intrapersonal biases; ego, overconfidence and emotion	"Egos of members of the team"
			"Thinking you have to be the font of all knowledge, and not seeking out advice from numerous sources to help you make that good decision"
			"Making decisions based on high emotion with a lack of perspective and rationality"
		Closed/fixed mindsets	"When working collaboratively the 'but we have always done it this way' group can be difficult"
			"Lack of a collective growth mindset/a fixed mindset within organisation—in some cases decisions are made by the organisation—the way its always been done, the way it's worked in the past, unwillingness for employees to push past these statements (to fall in line with more senior staff/ culture—even if failing) leads to poor decisions"

to align with the opinions of others, prioritize consensus over critical evaluation, and judge decisions based solely on their outcomes (7,20,33). Moreover, they reported interpersonal biases, specifically *egos*, *overconfidence*, *emotion*, and *closed/fixed mindsets* (Table 2), which can limit the consideration of alternative perspectives, lead to unwarranted confidence in judgments, and hinder the exploration of creative solutions (11,30). These biases align with those previously reported in the literature (2,19,22) and, as previously mentioned, emphasize the need for awareness and mitigation strategies, that can be acquired through formal training, to minimize their impact on decision quality.

Furthermore, subjects reported *environmental pressures* as obstacles to good decision-making. In such environments, psychological coping strategies become paramount for practitioners to handle these pressures while maintaining rationality (41). In addition, organizations can establish well-defined decision-making processes that, under time constraints, decentralize

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decision-making (46), empowering practitioners to take control of the decision-making process.

Practical Applications

In professional sports environments similar to those referenced by the subjects in the present study, findings suggest that performance support staff may benefit from education and training to enhance decision-making processes in their organizations. First, to create interest in decision-making education and enhance decision-making skills of practitioners, education on basic relatable psychological concepts may be an appropriate first step. More advance training may then be provided to overcome cognitive biases through evidence-based interventions. For example, Chang et al. (8) demonstrated that a single 45-minute training session in probabilistic-style thinking consistently improved forecasting accuracy. Second, to develop collaboration and overcome identified obstacles to team decision-making, evidence-based strategies, such as building psychological safety to overcome *poor communication* (10,34), may be useful. In addition, the regular inclusion of staff members outside of the normal performance support staff interdisciplinary team may mitigate against *conformity and group think* (13). Finally, to improve the monitoring of the process and outcome of decisions, strategies such as appointing devil's advocates as critical evaluators (43), and using checklists to ensure comprehensive information review (39), may enhance thoughtful updating of decisions. Although providing novel insights, this study has some limitations. Nonprobabilistic sampling (purposive and snowball sampling) was used to recruit subjects. Subsequently, findings may not generalize to all professional sport contexts. Second, although the authors attempted to recruit female subjects, only 12 of the 179 subjects were female. Thus, findings are gender biased. Future research in this domain may wish to adopt a more targeted strategy to engage female subjects. This is the first study to investigate performance support staff's perceptions relating to key facets of decision-making in professional sport. Although, performance support staff recognized the importance of decision-making in their roles and the value that making decisions in teams can bring, they also described the presence of *underdeveloped decision-making capabilities* within their organizations and significant obstacles to working within teams. Furthermore, performance support staff conveyed an absence of formal strategies and processes for decision-making at a personal and organizational level. When making decisions, performance support staff also faced several different *cognitive biases*, a lack of decision-making skills in others, and were challenged by *environmental pressures*. The evidence suggests that performance support staff may benefit from undertaking evidence-based training to improve decision-making skills. Moreover, implementing scientifically informed strategies to overcome obstacles may enhance decision-making processes at a personal, team, and organizational level.

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